

Adversarial Holdout for Reciprocal-Shell Contrasts: Partial Transfer and a High-Shell Obstruction

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Abstract

TPC-351 proposed the non-fitted reciprocal-shell contrast $\gamma_j = p_j^{-1} - r^{-1} \sum_k p_k^{-1}$ for the literal divisibility-mask defect in a prime-shell operator. We evaluate that same rule, without any refitting, on three disjoint origins and a new shell ladder. The exact zero-sum incidence identity and its induced-norm lower witness remain valid. On the frozen 144-row panel every reciprocal witness has positive response, but only 118/144 rows improve the balanced-step parent. The reciprocal response/defect ratio is 0.0801262572786–0.829632172143, compared with 0.099642909832–0.806767399067 for the parent; the reciprocal witness beats the coordinate baseline on 47/144 rows and reaches half the defect norm on 49/144. At $Q = 256$ its floor is lower than the parent. Thus the finite scale repair partially transfers but no uniform repair theorem is supported. No arithmetic L^2 estimate, fixed-power credit, or twin-prime result is claimed.

1 Question and literal defect

Let $I = \{o, o+1, \dots, o+M-1\}$ and let R_I, E_I denote restriction and zero extension. For a shell prime p , write $(P_p f)(n) = p \mathbf{1}_{p|n} f(n)$ and let K_s have entries

$$K_s(u, t) = \mathbf{1}_{u \neq t} \frac{H^{2s}}{(H^2 + (u-t)^2)^s}, \quad H = 66.$$

For fixed source signs ε_p , the physical and unmasked blocks are

$$A_I = \sum_p \varepsilon_p R_I P_p K_s P_p E_I, \quad T_I = R_I \left(\sum_p \varepsilon_p K_s \right) E_I, \quad D_I = A_I - T_I.$$

The two projection factors in A_I are retained literally. In particular, D_I is not replaced by a convolution and no mask term is discarded.

TPC-350 found that a fixed balanced shell contrast had positive finite responses but a poor high-shell floor. The present question is narrower: does one fixed, scale-aware rule transfer to a panel chosen independently of those responses?

2 Exact reciprocal-shell witness

For the ordered shell $p_0 < \dots < p_{r-1}$ define

$$\gamma_j = \frac{1}{p_j} - \frac{1}{r} \sum_{k=0}^{r-1} \frac{1}{p_k}, \quad c_I(t) = \sum_{j=0}^{r-1} \gamma_j \mathbf{1}_{p_j|t}. \quad (1)$$

The coefficients are rational and satisfy

$$\sum_j \gamma_j = \sum_j p_j^{-1} - r r^{-1} \sum_k p_k^{-1} = 0 \quad (2)$$

Table 1: TPC-352 paired finite holdout audit.

Quantity	Reciprocal	Balanced parent
Rows with positive response	144/144	144/144
Response/defect ratio, minimum	0.0801262572786	0.099642909832
Response/defect ratio, mean	0.397491684421	0.361474079935
Response/defect ratio, maximum	0.829632172143	0.806767399067
Rows reaching half defect	49/144	46/144
Rows beating coordinate baseline	47/144	30/144
Nondecreasing length series	22/48	22/48

Table 2: Scale-resolved response/defect ratios and half-defect counts.

Q	Reciprocal ratio			min	Parent ratio		Half defect	
	min	mean	max		mean	max	R	B
64	.575197	.695679	.829632	.472606	.632789	.806767	36	33
128	.162468	.419321	.782947	.110025	.372357	.698661	12	12
256	.080126	.225511	.432048	.099643	.215682	.366937	0	0
512	.136136	.249456	.544274	.134698	.225069	.508967	1	1

exactly. The rule is fixed by the shell alone; it does not depend on o , M , s , the source law, an entry of D_I , or an observed response.

Let $h_{p_j, I}$ be the interval vector $\mathbf{1}_{p_j|t}$. Multiply-divisible positions are intentionally retained, so linearity gives

$$c_I = \sum_j \gamma_j h_{p_j, I}, \quad \|D_I c_I\|_2^2 = \sum_{j,k} \gamma_j \gamma_k \langle D_I h_{p_j, I}, D_I h_{p_k, I} \rangle.$$

Consequently, whenever $c_I \neq 0$,

$$\|D_I\|_{2 \rightarrow 2} \geq \frac{\|D_I c_I\|_2}{\|c_I\|_2}. \quad (3)$$

Equations (2)–(3) are exact finite linear algebra. They do not assert cross-prime cancellation in the source variable.

3 Adversarial holdout protocol

The holdout was fixed before its responses were read:

$$o \in \{96097, 120097, 144097\}, \quad M \in \{256, 512, 1024\}, \quad Q \in \{64, 128, 256, 512\}, \\ s \in \{1, 2\}, \quad \varepsilon \in \{\text{all-plus, alternating-index}\}.$$

There are $3 \cdot 3 \cdot 4 \cdot 2 \cdot 2 = 144$ rows and 48 fixed origin–scale–law length series. On each row we compute the reciprocal witness and, on the identical matrix, the TPC-350 balanced parent (the first $\lfloor r/2 \rfloor$ shell coefficients are +1, the last are −1, and a middle coefficient is zero when r is odd). We also record the largest defect norm among mask-hit coordinate columns. No coefficient is selected after seeing any of these quantities.

The producer accumulates shells in ascending order. An independent checker accumulates them in reverse order, reconstructs all matrices, recomputes the spectral norms, and checks the exact rational anchor at $I = [193, 206]$, $Q = 4$, $s = 1$. A hostile checker applies eight protocol, certificate, anchor, census, claim, and parent-lock mutations.

4 Finite results

Table 1 gives the aggregate readout. The maximum reciprocal Gram replay discrepancy is $1.15463194561 \times 10^{-14}$.

The reciprocal response is larger than the balanced response on 118/144 rows. The gain range is 0.592696740468–1.47665316933 with mean 1.09769598704, so the comparison is plainly not uniform. The scale breakdown is:

The $Q = 256$ row is the decisive adversarial obstruction: reciprocal centering raises the mean slightly but lowers the minimum from the parent. At $Q = 512$ the improvement is again only partial. The data therefore support a scale-specific finite observation, not a scale-uniform repair.

4.1 Exact rational anchor

For $I = [193, 206]$ and the shell $(5, 7)$, equation (1) gives $(\gamma_0, \gamma_1) = (1/35, -1/35)$. The exact incidence vector is

$$c_I = \frac{1}{35}(0, 0, 1, -1, 0, 0, 0, 1, 0, 0, -1, 0, 1, 0), \quad \|c_I\|_2^2 = \frac{1}{245}.$$

With the literal rational defect at exponent one, the squared response is

$$\|D_I c_I\|_2^2 = \frac{2094105299985077452542621112101565891110487723654736503672553}{3198793259690951133635907806062856099910022277169847540500000}.$$

This anchor checks centering, multiply-divisible incidences, and the two-sided mask defect without relying on floating-point equality.

5 Validation and claim boundary

The producer and reverse-shell checker both pass in normal and optimized Python modes with byte-identical stdout and empty stderr. The stress suite rejects eight mutations. The local Bridge-B checker additionally locks the producer, checker, stress script, certificate, PDF, compile log, and bridge text before rerunning the finite checks.

The Session-named Route-A and Route-B evaluator files are absent from this checkout. Accordingly, the local result is fail-closed and is not an official evaluator pass. The exact statements proved here are finite coefficient balance, incidence linearity, Gram expansion, and the induced-norm test-vector inequality. The holdout census is numerical evidence only. In particular, this paper does not prove a source-uniform arithmetic L^2 estimate, a uniform masked-operator bound, a growing lower bound, a fixed-power saving, Route-B reassembly, or a twin-prime theorem.

6 Conclusion

The reciprocal-shell rule transfers as a useful but non-uniform finite witness: it improves most rows and both global half-defect and coordinate-baseline counts, yet its high-shell floor can be worse than the balanced parent. This is a genuine adversarial obstruction to promoting the TPC-351 repair into a uniform shell-scale principle. We therefore freeze this finite incidence branch and return the main route to the literal source-native masked arithmetic L^2 gate.

References

- [1] J. B. Conway, *A Course in Functional Analysis*, 2nd ed., Springer, 2000.